

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CL/EC/ME/ CE/EE/IT) SEM - I Examination, 2010-11

MA - 101 Engineering Mathematics

Date: 06.12.2010, Monday

Time: 10:30 a. m to 1:30 p. m.

Maximum Marks:70

Instructions:

1. Attempt all the questions in each section.
2. Assume suitable additional data if required.
3. Figures to the right side indicate the full marks.

SECTION-I

Q-1 Test the convergence of the following series:

[11]

(i) (a) $\frac{2}{1} + \frac{3}{4} + \frac{4}{9} + \dots + \frac{n+1}{n^2}$ (b) $\sum_{n=1}^{\infty} \frac{(n+\sqrt{n})^n}{3^n n^{n+1}}$ (c) $\sum_{n=1}^{\infty} \sqrt{n^4+1} - \sqrt{n^4-1}$

[9]

(ii) Prove that : $\frac{(\cos 2\theta - i \sin 2\theta)^5 (\cos 3\theta + i \sin 3\theta)^6}{(\cos 4\theta + i \sin 4\theta)^7 (\cos \theta - i \sin \theta)^8} = \cos 12\theta - i \sin 12\theta$.

[2]

Q-2 Attempt the following:

[12]

- (i) Find the roots common to the equations $x^4 + 1 = 0$ and $x^6 - i = 0$. $\text{cis}(\frac{7\pi}{12})$
- [5]
- (ii) Using De Moivre's theorem, prove that
- [4]
- (a) $\cos 5\theta = 5 \cos \theta - 20 \cos^3 \theta + 16 \cos^5 \theta$ (b) $\sin 5\theta = 5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta$

(iii) Prove that: $\tan \left\{ i \log \left(\frac{a-ib}{a+ib} \right) \right\} = \frac{2ab}{a^2 - b^2}$.

[3]

OR

Q-2 Attempt the following:

[12]

(i) Find the continued product of all values of $\left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)^{3/4}$

[5]

(ii) Expand $\sin^5 \theta \cos^2 \theta$ in a series of sines of multiple of θ

[4]

(iii) Find the modulus and argument of i^{1-i} mod = $e^{\pi/2}$

[3]

argument = $\frac{\pi}{2}$ or $\frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$

$$\begin{bmatrix} 1 & 1 & -2 & 0 \end{bmatrix}$$

- (ii) Using Gauss-Jordan method, Find the inverse of the matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$. [5]
- $A^{-1} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$
- (iii) Examine the consistency of the given system and if consistent solve them [4]

$$3x_1 + 2x_2 - x_3 = -15$$

$$5x_1 + 3x_2 + 2x_3 = 0$$

$$3x_1 + x_2 + 3x_3 = 11$$

$$-6x_1 - 4x_2 + 2x_3 = 30$$

OR

Q-3 Attempt the following:

[12]

- (i) Solve the system of linear equations: $x_1 + 3x_2 + 2x_3 = 0$, $2x_1 - x_2 + 3x_3 = 0$, [5]

$$3x_1 - 5x_2 + 4x_3 = 0, \quad x_1 + 17x_2 + 4x_3 = 0$$

- (ii) Find the rank & nullity of $A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$. [4]

- (iii) Discuss the convergence of the series [3]

$$\frac{1}{1^2+1} - \frac{2}{2^2+1} + \frac{3}{3^2+1} - \frac{4}{4^2+1} + \dots$$

SECTION-11

Q-4 Attempt the following:

[11]

- (a) Find the nth derivative of followings: (i) $y = \frac{x^4}{(x-1)(x-2)}$ (ii) $y = \sin^4 x$ [4+3]

- (b) If $y = a \cos(\log x) + b \sin(\log x)$, [4]

$$\text{then prove that } x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$$

- (b) (i) By using Lagrange's Mean Value Theorem, Find c if

$$f(x) = x(x-1)(x-2) \text{ in } \left[0, \frac{1}{2}\right]. \quad c = 1 - \sqrt{\frac{1}{2}}$$

[3+2]

(ii) Verify Cauchy's Mean Value Theorem for $f(x) = x^2$, $g(x) = x^3$ in $1 \leq x \leq 2$ $c = \frac{14}{9}$

- (c) If $u = \tan^{-1}\left(\frac{y}{x}\right)$, then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$.

[2]

OR

Q-5 Attempt the following:

- (i) If $V = \frac{1}{r}$ where $r^2 = x^2 + y^2 + z^2$ then prove that, $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$.

[12]

- (ii) If $u = \tan^{-1}\left(\frac{x^3 + y^3}{x - y}\right)$, then show that

[4]

(a) $x u_x + y u_y = \sin 2u$ & (b) $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 2 \sin u \cos 3u$.

- (iii) (a) If $f(x, y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2 + y^2}$ then prove that, $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + 2f(x, y) = 0$.

[2+2]

(b) Find $\frac{dy}{dx}$ if $(\cos x)^y = (\sin y)^x$.

Q-6 Attempt the following:

[12]

- (i) (a) If $u = \frac{y^2}{x}$, $v = \frac{x^2}{y}$, then Evaluate $J = \frac{\partial(x, y)}{\partial(u, v)}$ & $J' = \frac{\partial(u, v)}{\partial(x, y)}$ and hence verify $JJ' = 1$.

[3+2]

(b) The period of a simple pendulum is $T = 2\pi \sqrt{l/g}$. Find the maximum error T due to possible errors up to 1% in l & 2.5% in g .

- (ii) Examine the extreme values of $x^2 - 2xy + \frac{1}{3}y^3 - 3y$.

[3]

- (iii) (a) Find the maximum value of xyz subject to the restriction $x + y + z = a$.

[2+2]

(b) Find the equations of the tangent plane & the normal line to the surface

$2x^2 + y^2 + 2z = 3$ at the point $(2, 1, -3)$.

OR

$$x = \frac{y^2}{u}, \quad v = \frac{(y^2/u)^2}{y} = \frac{y^3}{u^2} \quad \Rightarrow \quad y^3 = u^2 v$$

$$y = (u^2 v)^{1/3}$$

$$x = \frac{u^{2/3} v^{1/3}}{u} = \frac{v^{1/3}}{u^{1/3}} = \left(\frac{v}{u}\right)^{1/3}$$

(ii) The temperature T at any point (x, y, z) in space is $T = kxyz^2$, where k is a constant. Find [3]

the highest temperature on the surface of the sphere $x^2 + y^2 + z^2 = a^2$.

(iii) Find the maximum and the minimum value of $2(x^2 - y^2) - x^4 + y^4$. [4]